

Bhavya Bansal

Computer Science + ML + CompFi @ Carnegie Mellon | Software Engineer | Creator

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🎓 EDUCATION

Expected May 2028

GPA: 4.0 / 4.0

Carnegie Mellon University | Pittsburgh, PA

B.S. in Computer Science, Minor in Machine Learning & Computational Finance

- Relevant coursework: 15-210 Parallel & Sequential DSA; 15-251 Theoretical CS; 15-213 Computer Systems; 15-451 Algorithms Design and Analysis (In Progress); 10-414 Deep Learning Systems (In Progress); 10-301; 21-370; 15-150, 15-281.

💼 PROFESSIONAL EXPERIENCE

May 2026 – Aug 2026

Ludhiana, Punjab, India

MasterTrust

Software Developer & Quantitative Researcher

- Engineered a full-stack execution system with concurrent order routing across users, real-time data pipeline, and a React frontend with strategy payoff visualization and live P&L tracking, improving workflow efficiency by 85% and cutting costs by 45%.
- Researched ML ensembles over weak predictive signals, improving mean accuracy from 51.3% individually to 57% in ensemble, with a custom backtesting framework supporting evaluation, performance benchmarking, and indicator refinement.
- Engineered a decision-tree-based strategy simulation pipeline to evaluate and stress-test them against historical data before live deployment.

Jun 2025 – Aug 2025

Ludhiana, Punjab, India

HNI Investor

Software Developer

- Developed a real-time data pipeline using VBS-driven process forking for live Excel reads, a WebSocket layer for price streaming, and a React frontend with interactive charting and dynamic filtering.
- Engineered an audio alert system that triggers on threshold-based price changes, while ensuring separate routing of the notification layer from the data pipeline for responsive and low-latency UX.

🔧 PROJECTS

May 2025 – Jul 2025

AI Study Buddy

Full-Stack | React, Tailwind, Flask, OpenAI, Judge0(RapidAPI)

- Built a full-stack coding practice platform with an LLM tutor and a Flask backend that merges user submissions into language-specific test harnesses and routes them to Judge0 API.
- Implemented a lightweight client-side problem index for instant filtering, and enforced structured JSON contracts between the AI layer and UI via system prompts.

Apr 2023 – Aug 2025

My Med Tracker

Mobile | React Native, Google Cloud Computing, Firebase, RSA, JS

- Built a React Native app with Firebase-backed prescription management, RSA-encrypted patient data, and role-separated CRUD flows for doctors and patients.
- Provided automated medication reminders to 1,500+ patients in collaboration with 10+ verified doctors across 2 different hospitals in Ludhiana.

💡 SKILLS

Technical: Python, C/C++, SML, Java, JS, SQL • PyTorch, React, Flask, React Native, Firebase, AWS.

Domains: Full-stack development, systems programming, cloud infrastructure, SaaS.

Languages: English (native), Hindi (native), Punjabi (proficient).